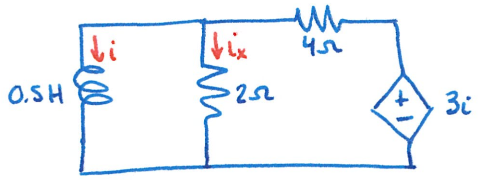
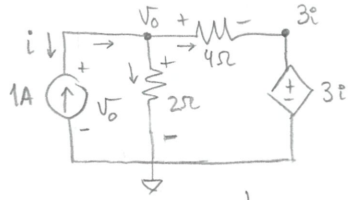
EXERCISE 2 (Natural response RL)

Find i and i_x if $i(0) = 10$ A



* First, calculate R_{th}



$R_{th} = \frac{V_0}{i_0}$

$i = -1$ A

Nodal analysis:

$$1 - \frac{V_0}{2} - \frac{V_0 - 3i}{4} = 0$$

$$1 - \frac{V_0}{2} - \frac{V_0}{4} + \frac{3}{4}\underbrace{i}_{-1} = 0 \Rightarrow -\frac{3}{4}V_0 = -1 + 3/4$$

$$-3/4\,V_0 = -\frac{1}{4}$$

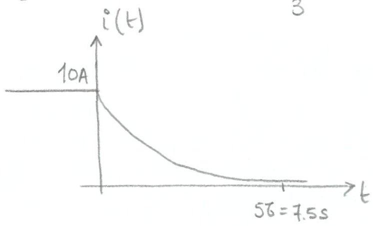
$$V_0 = \frac{1}{3}\,V$$

$$R_{th} = \frac{V_0}{i_0} = \frac{1/3\,V}{1A} = 1/3\,\Omega$$

* The calculate time constant and natural response

$$\tau = \frac{L}{R_{th}} = \frac{1/2\,H}{1/3\,\Omega} = \frac{3}{2}\,s$$

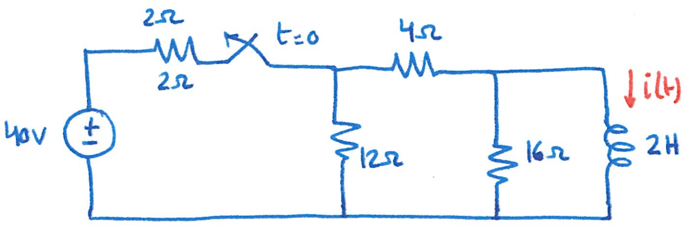
$$i(t) = I_0 e^{-t/\tau} = 10 e^{-\frac{2t}{3}}\,A \Longrightarrow$$



$$i_x(t) = \frac{V_L(t)}{2} = \frac{1}{2}V_L(t) = \frac{1}{2}L\frac{di}{dt} = \frac{1}{2}\times\frac{1}{2}\times 10\times\left(\frac{-2}{3}\right)e^{-2t/3} = -\frac{5}{3}e^{-2t/3}\,A$$

EXERCISE 3 (Natural response RL)

Find $i(t)$ for $t>0$

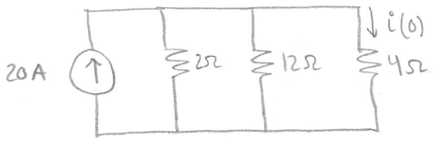


* First, find the initial conditions $(t<0)$

Inductor in DC steady state behaves as short circuit

16Ω resistor is bypassed by short circuit

Source transformation

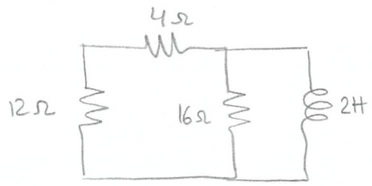


$$i(0) = 20\,\frac{2\|12\|4}{4} = \frac{20}{4}\times\frac{12}{10} = 6A$$

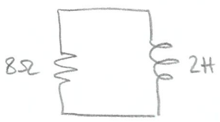
current divider

$$2\|12\|4 = \frac{1}{\frac{1}{2}+\frac{1}{12}+\frac{1}{4}} = \frac{1}{\frac{6+1+3}{12}} = \frac{12}{10}\Omega$$

* Then, update circuit and calculate natural response:



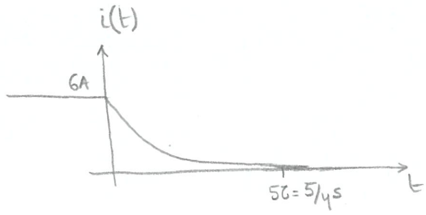
simplify



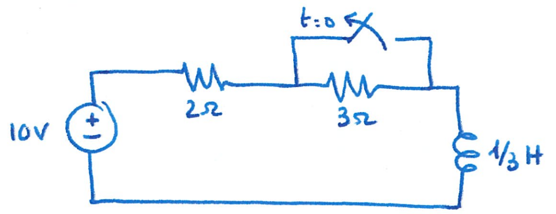
$$R_{eq} = R_{th} = (12+4)\|16 = 8\Omega$$

$$\tau = \frac{L}{R_{th}} = \frac{2H}{8\Omega} = \frac{1}{4}s$$

$$i(t) = I_0\, e^{-t/\tau} = \underline{\underline{6\,e^{-4t}\ A}}$$

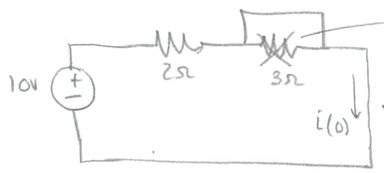


EXERCISE 4 (Step response RL)



Find $i(t)$ for $t \geq 0$

* First, calculate initial conditions ($t \leq 0$)



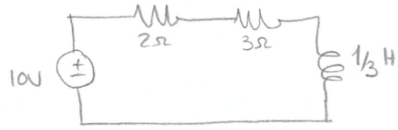
3Ω resistor is bypassed by short circuit

inductor in DC steady state behaves as short circuit

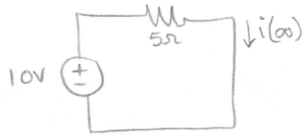
$$i(0) = \frac{10V}{2\Omega} = 5A$$

* Then, update circuit and calculate step response:

$$i(t) = i(\infty) + (i(0) - i(\infty))e^{-t/\tau}$$



We calculate $i(\infty)$ ⟶ DC steady state again ⟶ L is short circuit



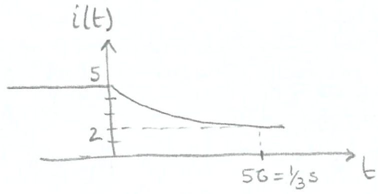
$$i(\infty) = \frac{10V}{5\Omega} = 2A$$

We calculate τ:

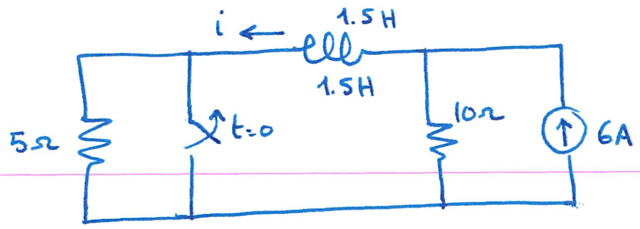
$$\tau = \frac{L}{R} = \frac{1/3\ H}{5\Omega} = 1/15\ s$$

$R_{th} = R$

$$i(t) = i(\infty) + (i(0) - i(\infty))e^{-t/\tau} = 2 + (5-2)e^{-15t} = \underline{\underline{2 + 3e^{-15t}\ A}}$$



EXERCISE 5 (Step response RL)



* First, calculate initial conditions $i(0)$ $(t<0)$

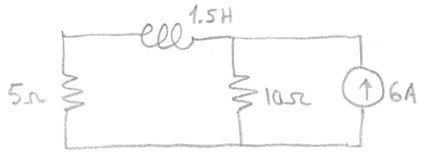
→ L in DC steady state ⇒ short circuit

→ $i(0) = 6A$

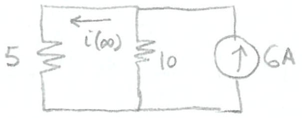
5Ω resistor is bypassed by short circuit

10Ω resistor is bypassed by short circuit

* Then, update circuit and calculate step response : $i(t) = i(\infty) - (i(0) - i(\infty))e^{-t/\tau}$



We calculate $i(\infty)$ → DC steady state again → L is short circuit

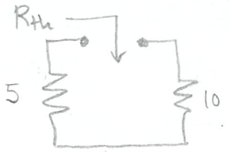


$i(\infty) = 6\,\frac{10\|5}{5} = \frac{6}{5} \times \frac{10}{3} = 4A$

current divider

$10\|5 = \frac{50}{15} = \frac{10}{3}\,\Omega$

We calculate $\tau = \frac{L}{R_{th}}$:



$R_{th} = 5 + 10 = 15\,\Omega$

$\tau = \frac{L}{R_{th}} = \frac{1.5H}{15\,\Omega} = 0.1\,s$

$$i(t) = i(\infty) + (i(0) - i(\infty))\,e^{-t/\tau} = 4 + (6-4)\,e^{-t/0.1} = 4 + 2e^{-10t}$$